

Advanced (Habanero)

- Q1.** What is the graph of the system of inequalities $x \geq 2$ and $y \leq 3$?
- (A) A line at $x = 2$ and a line at $y = 3$
(B) A line at $x = 2$ and a line at $y = -3$
(C) A line at $x = -2$ and a line at $y = 3$
(D) A line at $x = 2$ and a line at $y = 0$
(E) A line at $x = 0$ and a line at $y = 3$
- Q2.** The system of inequalities $x + y \geq 4$ and $2x - 2y \leq 4$ has a solution region that is
- (A) Unbounded
(B) Bounded
(C) Empty
(D) A single point
(E) A line
- Q3.** The system of inequalities $y \geq x - 2$ and $y \leq -x + 2$ has a solution region that is
- (A) A triangle
(B) A rectangle
(C) A trapezoid
(D) A parallelogram
(E) A hexagon
- Q4.** The system of inequalities $x \geq 0$, $y \geq 0$, and $x + y \leq 4$ represents a
- (A) Triangle
(B) Quadrilateral
(C) Pentagon
(D) Hexagon
- (E) Heptagon
- Q5.** What is the corner point of the solution region for the system $x \geq 0$, $y \geq 0$, $x + y \leq 4$, and $x - y \leq 2$?
- (A) (0, 0)
(B) (2, 2)
(C) (4, 0)
(D) (0, 4)
(E) (2, 0)
- Q6.** The system of inequalities $y \geq x - 3$ and $y \leq 2x + 1$ has a solution region that is
- (A) Above the line $y = x - 3$ and below the line $y = 2x + 1$
(B) Below the line $y = x - 3$ and above the line $y = 2x + 1$
(C) On the line $y = x - 3$ and on the line $y = 2x + 1$
(D) Above the line $y = x - 3$ and on the line $y = 2x + 1$
(E) On the line $y = x - 3$ and below the line $y = 2x + 1$
- Q7.** Which system of inequalities has the solution region shown?
- (A) $x \geq 0$, $y \geq 0$, $x + y \leq 6$, $x - y \leq 2$
(B) $x \geq 0$, $y \geq 0$, $x + y \leq 4$, $x - y \leq 2$
(C) $x \geq 0$, $y \geq 0$, $x + y \leq 8$, $x - y \leq 4$
(D) $x \geq 0$, $y \geq 0$, $x + y \leq 10$, $x - y \leq 6$
(E) $x \geq 0$, $y \geq 0$, $x + y \leq 12$, $x - y \leq 8$
- Q8.** The system $x + y \geq 2$ and $x - y \leq 4$ can be represented as
- (A) A line and a ray
(B) Two intersecting lines
(C) Two parallel lines
(D) A line and a curve
(E) A curve and a ray

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities $x + y \leq 6$, $x - y \geq -2$, $x \geq 0$, $y \geq 0$. Identify the corner points of the solution region.
- Q10.** Graph the system of inequalities $y \geq x - 1$, $y \leq 3x + 2$, $x \geq 0$, $y \geq 0$. Find the area of the solution region.

Chef's Tips

- Tip 1: Make sure students understand the concept of corner points and feasible regions.
- Tip 2: Have students use graph paper to ensure accurate graphs.
- Tip 3: Consider having students work in pairs to facilitate discussion and learning.